

Approach Overview

We built a probability of default model using 23 raw fields and engineered behavioural features, trained with LightGBM, calibrated per-fold, and post-processed with a base-rate alignment step.

Data Cleaning and Preprocessing

- No missing values and duplicate rows were found (excluding client_id)
- Undocumented category codes for EDUCATION (0,5,6) were put into another bucket (4) and for MARRIAGE (0) were put into (3). These weren't dropped, since the test set contains the same undocumented codes.

Feature Engineering

Additional features were produced on top of the raw 23 columns to summarise the overall pattern of a customer's credit behaviour across multiple months.

Group	Features	Explanation
Utilization	UTIL_1-UTIL_6, UTIL_AVG	Monthly bill balance relative to credit limit and average usage
Payment coverage	PAY_RATIO_1-PAY_RATIO_5, PAYRATIO_AVG	Payment amount relative to the corresponding previous bill
Delinquency	MAX_DELAY, SUM_DELAYS, NUM_MONTHS LATE	Severity, cumulative amount and frequency of late repayment status
Trend	WORSENING, BILL_TREND	Recent vs. Historical trajectory
Spending level	HEADROOM, BILL_STD, PAYAMT_STD	Unused credit, and volatility of bills across 6 months
Categorical	SEX, EDUCATION, MARRIAGE	Nominal codes per the data dictionary, not an ordered scale

Validation Strategy

10-fold stratified cross-validation on the training set. Within each fold's training portion, a further 15% is held out purely for isotonic calibration. LightGBM training uses early stopping against the fold's validation set. Out-of-fold predictions across all 10 folds are combined to report an overall cross-validated log loss.

Models tested and final model selection

We tested several modelling approaches, including Logistic Regression, XGBoost, and an XGBoost–LightGBM blend, with different levels of tuning and feature engineering. After comparing their performance, the tuned LightGBM model with isotonic calibration, pseudo-labelling, and base-rate alignment was selected as the final model. It achieved a leaderboard log loss of 0.40873, making it the best-performing approach tested.

Ensembling

- **Pseudo-labeling:** An initial LightGBM model is trained on the full training set and used to generate predictions on the test set. Test rows with predicted probability more than 0.95 or less than 0.05 are added back into the training set with a hard pseudo-label (1 or 0 respectively), and the final 10-fold model is trained on this augmented set.
- **Calibration:** Isotonic regression per fold, fit on a held-out calibration split.
- **Clipping:** Final predictions clipped to [0.015, 0.985] to bound worst-case log loss exposure.
- **Base-rate alignment:** The final probabilities were adjusted in logit space so that their average matched the known training default rate of 22.12%

Key Results

- Best cross-validated, leaderboard log loss was 0.40873.
- Repayment-history features (PAY_0, MAX_DELAY, SUM_DELAYS) dominate feature importance while demographic fields contribute comparatively little, consistent with EDA.